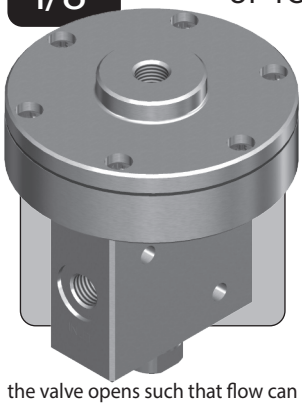
B1-12

STOP VALVE

1/8" UP TO 1379 BAR 20,000 PSI



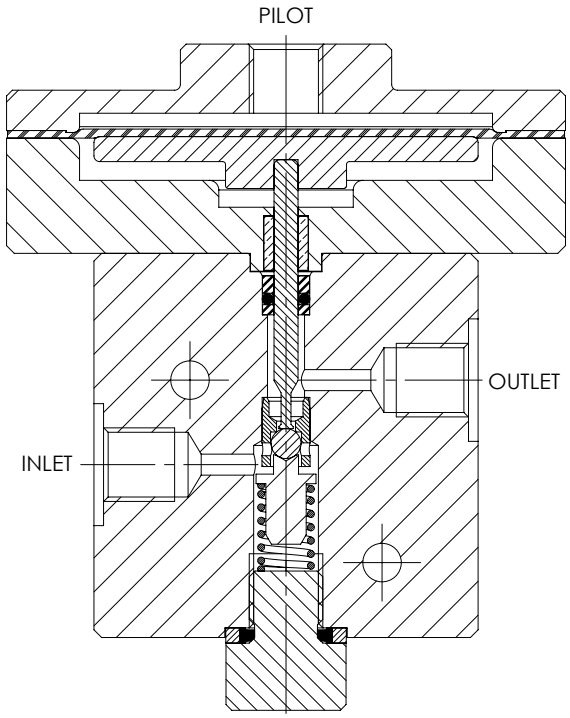
The B1-12 is a 1/8" nominal bore pilot operated check valve. With no pilot pressure applied, the valve will block flow from the inlet to the outlet port, while allowing flow from outlet to inlet subject to the differential pressure exceeding the cracking pressure.

The valve is fitted as standard with a pneumatic actuator. Once the actuator is pressurised, the valve opens such that flow can pass in both directions.

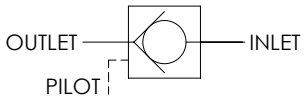
- Pilot operated check valve
- Low pressure diaphragm actuator, ideal for pneumatic actuation
- Low leakage
- Suitable for use with mineral oil and water glycols
- Vent chamber prevents mixing of main line and pilot fluid
- Various porting options available
- Pilot pressure to start opening may be calculated using the following formulae for the 10K and 20K versions

20K $\frac{\text{Inlet Port Pressure + Cracking Pressure + (0.62 x Outlet Port Pressure}}{\text{Pilot Ratio}}$

10K $\frac{\text{Inlet Port Pressure + Cracking Pressure}}{\text{Pilot Ratio}}$



Specifications

BASIC MODEL NUMBER	B1-12
SYMBOL	
MAX WORKING PRESSURE	1379 bar (20,000 psi)
MAX PILOT PRESSURE	10.3 bar (150 psi)
CV (FLOW CAPACITY)	10K - TBA 20K - TBA
PILOT RATIO	10K - 250:1 20K - 400:1
CRACKING PRESSURE	10K - 33 bar (480 psi) 20K - 56 bar (815 psi)
FLUIDS	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/8"
WEIGHT	1.1 kg (2.4 lb)

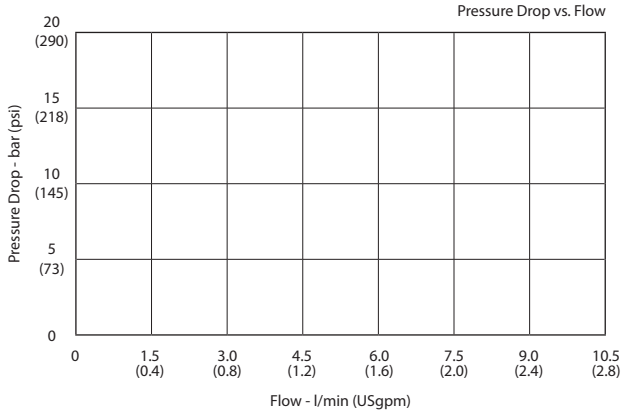
Specifications may change without notice

Materials

Externally Exposed Parts: 316, 303 and 431 stainless steel. Zinc plated screws.
Internally Wetted Parts: 316, 302, 303, 431 and 17/4 PH stainless steel, silicon nitride and PTFE. The actuator contains a nitrile diaphragm.
The standard valve is designed for use with mineral oil and water glycols with an air pilot, but may also be used with a wide variety of media compatible with the materials of construction. Contact us for advice.

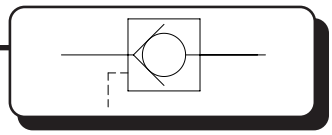
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

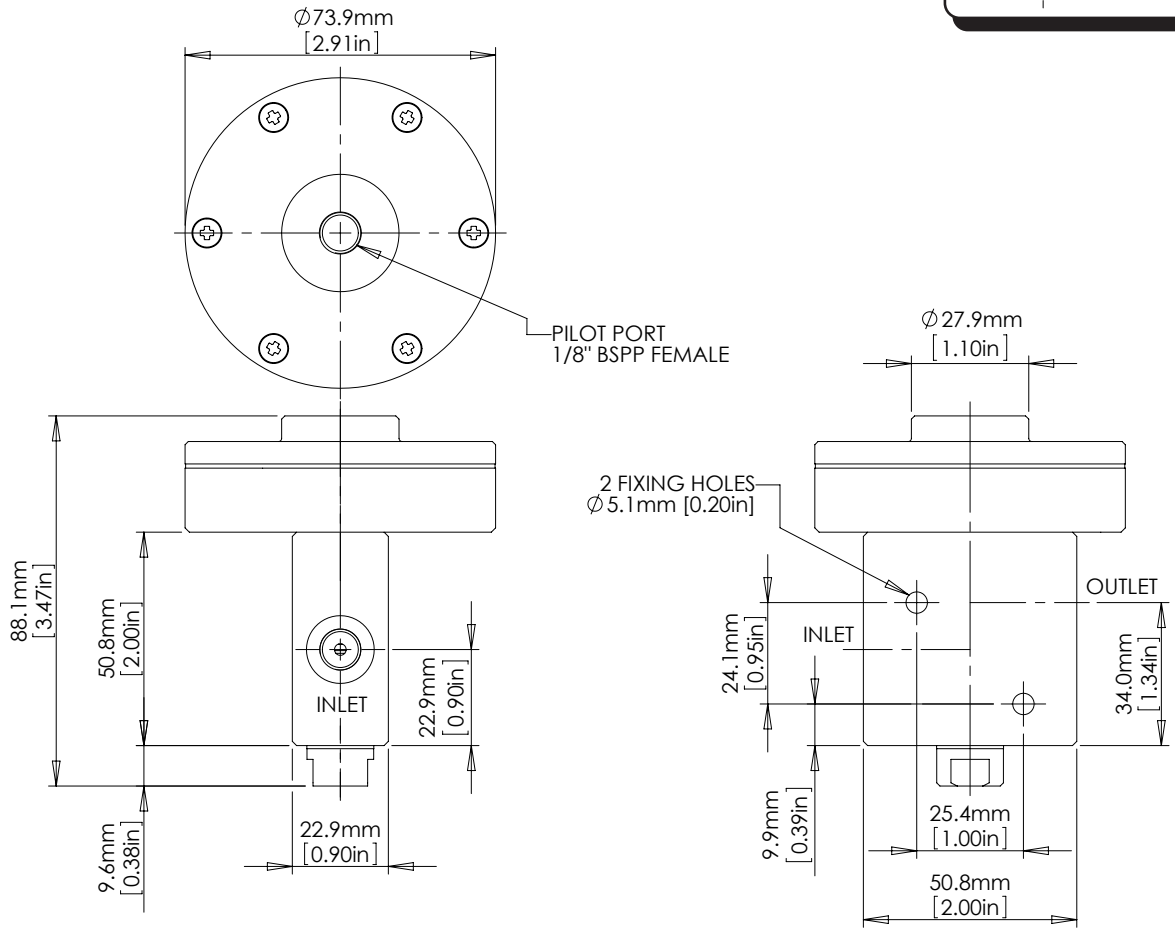
Typical Performance



Typical performance based on water
Graph shows the valve fully open

Installation Diagram





Product Selector

B1-12 P - NBR - 10K

VALVE TYPE	PORTING OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
B1-12	P 1/8" BSPP female 56AEK 3/8" OD medium pressure tube port ⚠ The pressure rating of connecting pipework must be considered See installation drawing above for pilot port type and size	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -30˚C to +105˚C Note that all seal options use a Nitrile diaphragm	10K 690 bar (10,000 psi) Liquid 20K 1379 bar (20,000 psi) Liquid

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK